

Supplementary Information

Interpolation versus generalization in machine learning prediction of micropollutant adsorption on layered double hydroxides

[AUTHORS]

Text S1 Data cleaning rules

All rules are implemented in `ldh_pipeline.load_data` and applied identically in every notebook. (1) Column names are stripped of surrounding whitespace. (2) Categorical labels are normalized by lowercasing, collapsing internal whitespace and hyphen spacing, and stripping trailing punctuation, after which known synonyms are merged (e.g., *co-precipitation or aging*, *co-precipitation* $\rightarrow$ *Co-precipitation*; *reconstruction or memory effect* $\rightarrow$ *Reconstruction*). (3) The removal ratio is converted to percent units by a range-aware rule: if the column maximum is ≤ 1.5 the values are multiplied by 100; an assertion then checks that all values lie in $[0, 100]$, which makes the conversion idempotent under re-execution. (4) Records identical in every descriptor are reduced to their first occurrence (83 records removed). (5) Missing numeric values are imputed inside each model pipeline by a five-neighbor KNN imputer computed in z-score space and mapped back to physical units; columns with no observed value in a training fold fall back to the fold mean. The source-study identifier is retained for grouping only and never enters the feature matrix.

Text S2 Hyperparameter search spaces and selected values

Tree-ensemble hyperparameters were selected by Optuna (TPE sampler, 30 trials, seeded) on cross-validation folds of the training data of the reference split; grid searches used the same folds. Selected values are read directly from the saved model pipelines (`Model/interp_row_42/saved_model`).

Text S3 Per-model results under both tasks

Text S4 Additional figures

Table S1: Hyperparameter search spaces and selected values

Model	Search space	Selected
HistGB	max_iter 100–600; lr 10^{-4} –1 (log); depth 3–15; leaf 3–9	max_iter 484; lr 0.089; c
XGBoost	n_estimators 100–600; lr 10^{-4} –1 (log); depth 3–15; child weight 3–9	495 trees; lr 0.114; dept
RF	n_estimators 100–600; depth 3–15; leaf 3–9	347 trees; depth 13; lea
Cubist	committees 1–9; neighbors none, 1–8	committees 9; neighbors
KNN	k 1–9	$k = 1$
ANN	fixed: 1 hidden layer, 32 ReLU units, Adam, 2000 iterations	—
LR	no hyperparameters	—

Table S2: Leave-one-study-out per-study MAE distribution (RR%), 59 studies, fixed hyperparameters

Model	Median	Q25	Q75
Cubist	14.1	4.7	22.2
KNN	16.0	6.5	30.3
HistGB	16.4	4.7	25.8
RF	16.8	6.1	25.9
ANN	18.1	11.5	38.6
XGB	18.3	3.8	25.4
LR	28.4	18.4	34.8
Global-mean baseline	24.2	—	—
Per-pollutant-mean baseline	25.1	—	—

Table S3: Wall-clock training and prediction time on the reference split (seconds; single measurement, indicative only)

Model	Fit	Predict
KNN	0.05	0.04
LR	0.14	0.05
Cubist	0.22	0.04
RF	0.37	0.09
ANN	0.67	0.01
HistGB	0.86	0.08
XGBoost	10.1	0.03

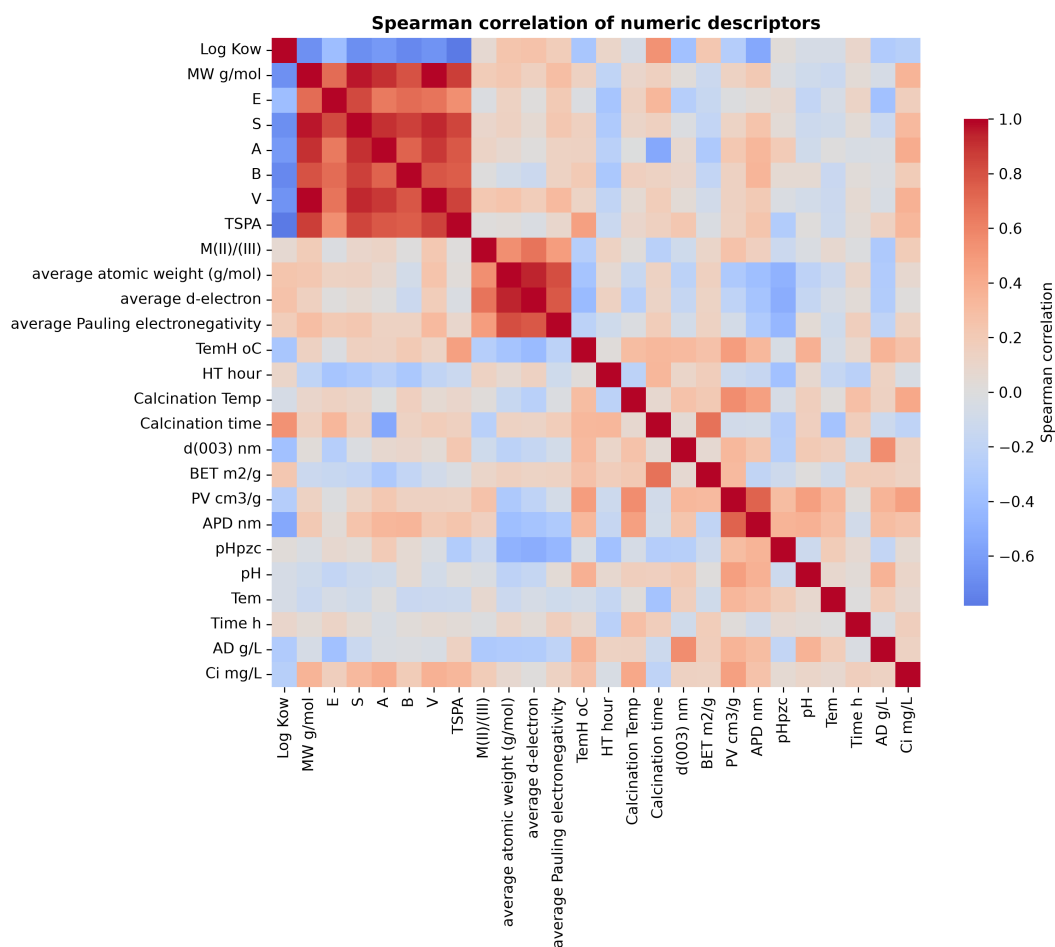


Figure S1: Spearman correlation of the numeric descriptors

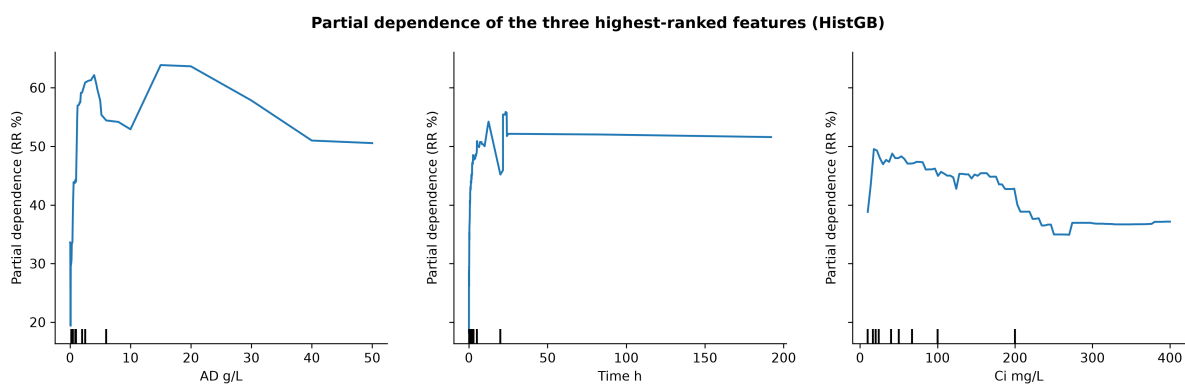


Figure S2: Partial dependence of the three highest-ranked features (HistGB)

SHAP dependence of $\text{pH}-\text{pH}_{\text{pzc}}$, colored by Log Kow (HistGB, test set)

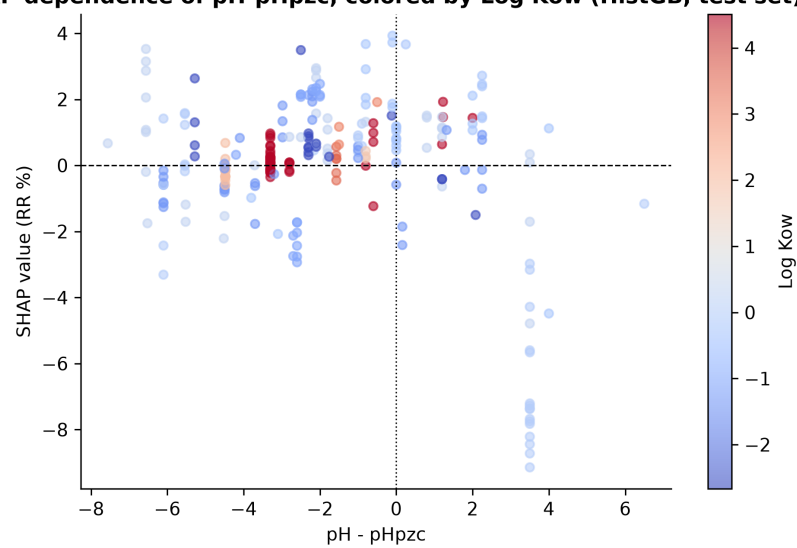


Figure S3: SHAP dependence of $\text{pH}-\text{pH}_{\text{pzc}}$ colored by $\log K_{\text{ow}}$ (HistGB with engineered features, hold-out test set)

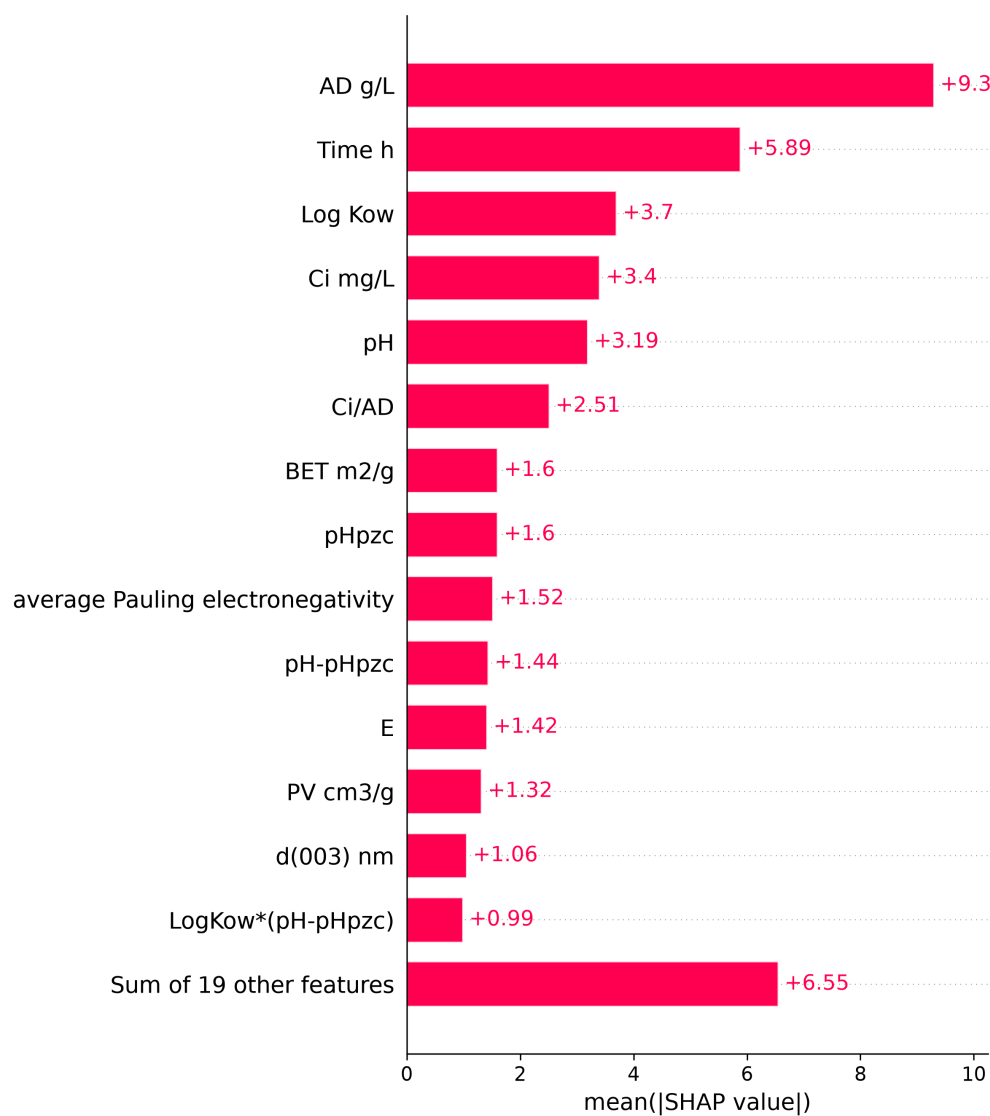


Figure S4: Global SHAP importance, HistGB with engineered features